LAB SHEET 05: FORM VALIDATION ARCHITECTURE & CONTROLLED INPUTS IN REACT

Aim

Create a functional, responsive simple Login Form module using controlled component inputs in React JS.

Task 5.1 — Dynamic Regex Validation

Input / Source Code import {

useState } from "react";

function LoginForm() {

const [email, setEmail] = useState("");

const [password, setPassword] = useState("");

const [error, setError] = useState("");

const handleSubmit = (e) => {

e.preventDefault();

const emailRegex = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;

const passwordRegex = /^(?=.*[A-Za-z])(?=.*\d).{6,}$/;

if (!emailRegex.test(email))

setError("Invalid email format.");

else if (!passwordRegex.test(password))

setError("Password must contain letters, numbers and 6+ characters.");

else {

setError("");

alert("Login Successful!");

}

};

return (

<h2>LoginForm</h2>

<input

type="email"

placeholder="Enter Email"

value={email}

```
onChange={(e) => setEmail(e.target.value)}
    />
    <br /><br />
    <input
      type="password"
 placeholder="Enter Password"
value={password}
onChange={(e) => setPassword(e.target.value)}
    />
    <br /><br />
    <button onClick={handleSubmit}>Login</button>
    <p>{error}</p>
   </div>
  );
}
export default LoginForm;
```

Output

Login Form

[Enter Email]

[Enter Password]

[Login]

Invalid email format.

For valid details:

Login Successful!

Task 5.2 — Password Strength Checker

Input / Source Code

```
import { useState } from "react";
function PasswordStrength() {
const [password, setPassword] = useState("");
```

```
 const strength =
    password.length < 6 ? "Weak" :
   password.length < 10 ? "Medium" : "Strong";
 return (
   <div>
     <h2>Password Strength Checker</h2>
     <input
       type="password"
 placeholder="Enter Password"
value={password}
onChange={(e) => setPassword(e.target.value)}
     />
     <p>Password Strength: {strength}</p>
     <progress
value={Math.min(password.length, 10)}
max="10"
     />
   </div>
  );
}
export default PasswordStrength;
```

Output

Password Strength Checker

[Enter Password]

Password Strength: Weak

[██░░░░░░░░]

For a stronger password:

Password Strength: Medium

[██████░░░░]

Password Strength: Strong

[██████████]

Task 5.3 — Multi-Step User Onboarding Form

Input / Source Code import { useState } from

"react";

function OnboardingForm() {

const [step, setStep] = useState(1);

const [data, setData] = useState({

name: "",

email: "",

age: ""

});

const updateData = (e) => setData({ ...data,

[e.target.name]: e.target.value });

return (

<h2>User Onboarding</h2>

{step === 1 && (

<>

<h3>Step 1: Personal Information</h3>

<input

name="name"

placeholder="Enter Name"

value={data.name}

onChange={updateData}

/>

<button onClick={() => setStep(2)}>Next</button>

</>

)}

{step === 2 && (

```
      <>
        <h3>Step 2: Contact Information</h3>
        <input
          name="email"
placeholder="Enter Email"
value={data.email}
onChange={updateData}
        />
        <br /><br />
        <button onClick={() => setStep(1)}>Previous</button>
        <button onClick={() => setStep(3)}>Next</button>
      </>
    )}
    {step === 3 && (
      <>
        <h3>Step 3: Additional Information</h3>
        <input
name="age"
placeholder="Enter Age"
value={data.age}
onChange={updateData}
        />
        <br /><br />
        <button onClick={() => setStep(2)}>Previous</button>
        <button onClick={() => alert("Onboarding Completed!")}>
          Submit
        </button>
      </>
    )}
  </div>
```

```
  );
}
export default OnboardingForm;
```

Output

Step 1:

User Onboarding

Step 1: Personal Information

[Enter Name]

[Next]

Step 2:

User Onboarding

Step 2: Contact Information

[Enter Email]

[Previous] [Next]

Step 3:

User Onboarding

Step 3: Additional Information

[Enter Age]

[Previous] [Submit] After

clicking Submit:

Onboarding Completed!